

Assignment 01 [SKZ | Summer 2026]

Total points 80/100 ?

Assignment #1 on Application Layer (HTTP)

The respondent's email (**md.rimon.hossain2@g.bracu.ac.bd**) was recorded on submission of this form.

Student ID *

23201644

✓ What do 1 and 2 represent below? *

5/5



- ☐ Status Line and Header Line
- ☐ Feeder Line and Header Line
- ☐ Status Line and Feeder Line
- ☒ Request Line and Header Line



✓ A web page is a collection of objects, and those objects can be (choose 2)

*5/5

☐ email

☒ html text

✓

☐ grid

☐ blossom

☒ png image

✓

✓ When you access www.yahoo.com's web page, for each object of the web page, a tcp connection is opened. The HTTP type is then *5/5

☐ Consistent

☐ Persistent

☒ Non Persistent

✓

☐ Pipelining

☐ Non Consistent

✓ In HTTP, _____ server is a computer that keeps copies of web pages of most recent requests. *5/5

☐ an email

☒ a proxy

☐ a DNS

☐ a remote



✗ Suppose a web page consist of 5 objects in total. Considering that the connection type is non-persistent, If one RTT = 3ms, calculate the **total RTT in ms** required to download the webpage. *.../5

5*(2*RTT) = (5*6) = 30 ms



Correct answers

30

30 ms

30ms



✗ Suppose a web page consist of 5 objects in total. Each object is 10 MB. *.../5
The web server's transmission speed is 100Mbps. Calculate the file transmission time in **ms** for all the objects. Just input the number.

4,000

✗

Correct answers

4000

4000 ms

4000ms

✗ Ruha has requested a webpage using a **non-persistent HTTP** connection. *0/5
The webpage has **34 objects, excluding the base HTML** file, and each object is **4 MB**. It takes **35 ms** to send a **TCP request** from her device to the server and come back. Also, the **HTTP request and response** for each object is **30 ms**. The server has a speed of **64 Mbps**.
Calculate the total RTT required in **ms** for all objects. **Just input the number value.**

2,275

✗

Correct answers

2275

2275 ms

2275ms

✗ Ruha has requested a webpage using a **non-persistent HTTP** connection. *0/5
The webpage has **34 objects, excluding the base HTML** file, and each object is **4 MB**. It takes **35 ms** to send a **TCP request** from her device to the server and come back. Also, the **HTTP request and response** for each object is **30 ms**. The server has a speed of **64 Mbps**.
Calculate the total file transmission time in ms for all webpage objects. **Just input the number value.**

17,000



Correct answers

17500

17500 ms

17500ms

17.5

✓ When you send a HTTP request using Chrome browser, which method is being used? *5/5

☐ PUT

☐ LOAD

☒ GET

☐ POST



✓ You are registering yourself at Daraz.com. Which method is used to to upload your personal information at the Daraz server? *5/5

- ☐ GET
- ☐ PUT
- ☒ POST
- ☐ HEAD



✓ HTTP uses conditional GET so that * 5/5

- ☐ Web cache can ensure uncorrupted file
- ☒ Web chache can ensure up to date files
- ☐ Web cache can ensure files at a faster rate
- ☐ Web cache can ensure files at a slower rate



✓ Suppose Brac university's internal bandwidth is 200 Mbps, and the access link to the BracNet is 20 Mbps. On an average 40 requests of the size 1 Mbits are sent per second. Now what is the LAN delay and the Access delay? *5/5

- ☐ 0.25 and 2
- ☐ 0.10 and 1
- ☐ 0.20 and 1
- ☒ 0.20 and 2



✓ Which HTTP method leaves out the requested object in the HTTP response message?

*5/5

- ☐ GET
- ☐ PUT
- ☐ POST
- ☒ HEAD



✓ Asif has requested a webpage using a persistent HTTP connection. The webpage has **18 objects**, including the base HTML file. The total RTT required to retrieve all objects is **294 ms**. The TCP request takes **12 ms** (one-way). The total file transmission time for all webpage objects is **24,000 ms**.

*5/5

(i) Calculate the **total HTTP RTT** required for all the webpage objects.

- ☐ 250
- ☒ 270
- ☐ 284
- ☐ 312



✓ For the same question above, (ii) Calculate the server speed in Mbps. * 5/5

- ☐ 84
- ☐ 64
- ☒ 72
- ☐ 82



✓ HTTP is stateless, what does it mean? * 5/5

- ☐ All user requests are being tracked
- ☐ Receiver must retain session state from previous requests.
- ☒ Request to a Web server, the response you'll get will not depend on earlier requests.
- ☐ Requests are sent using cookies



✓ What are true regarding Cookies? * 5/5

- ☐ It is not stored in web browsers.
- ☐ It is generated by the web browsers.
- ☒ It is generated by the web site server.
- ☐ It is sent from Root servers to all servers.



✓ Given, you are contacting a server which has the capacity to send 50 objects per second of size 1Mb each. (a) What would be the % utilization of the access link if the access link speed is 120Mbps. *5/5

☐ 50.20%

☒ 41.67%

☐ 37.85%

☐ 46.35%



✓ HTTPS stands for Hypertext Transfer Protocol over TLS. 5/5

☒ True

☐ False



✓ How does HTTPS ensure security? 5/5

☐ By resending data when they are corrupted

☐ By correcting errors in the data packet

☒ By encrypting data while in transfer

☐ By dropping data



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